

# Jilei Lin

Ph.D. in Statistics, The George Washington University  
Arlington, VA, U.S.

jileil2@gwmail.gwu.edu  
ORCID: 0000-0002-7147-9851  
Google Scholar | GitHub

## Research Profile

I develop statistical methodology for heterogeneous, spatially dependent, and censored data, with applications in medicine, environmental health, spatial transcriptomics, and social science.

**Core methods:** Quantile regression; distributed inference; spatial analysis; dimension reduction; survival analysis; missing data analysis; longitudinal data analysis; transfer learning.

## Education

**2021–2026**      **Ph.D. in Statistics**, The George Washington University, Washington, D.C.

Dissertation: *Quantile Regression for Spatial Data and Censored Data*.

Advisors: [Huixia Judy Wang](#) and [Feifang Hu](#).

**2019–2021**      **M.S. in Statistics**, University of Illinois Urbana-Champaign, IL.

**2015–2019**      **B.S. in Finance**, *summa cum laude*, Kean University, Union, NJ.

## Publications

### Published and Accepted

- [1] **Smoothed Quantile Regression for Spatial Data.**  
[Lin, J.](#), [Wang, H.](#), [Wang, L.](#), and [Kim, M.](#) **Accepted**; *Journal of Computational and Graphical Statistics (JCGS)*. Software: [jileil2/SQBiT](#).
- [2] **Hybrid Supervised-unsupervised Modeling for Post-hurricane Private Well Contamination Risk Score Using Empirical Validation and Community Groundtruthing.**  
[Lin, J.<sup>†</sup>](#), [Zhang, J.](#), [Wei, E.](#), [Shea, K.](#), [Wang, H. J.](#), [Liddie, J. M.](#), [Hernandez, E.](#), [Norford, C.](#), and [Hu, X. C.](#) **Accepted**; *GeoHealth*. doi:10.1029/2026GH001858.
- [3] **Minimizing Post-shock Forecasting Error through Aggregation of Outside Information.**  
[Lin, J.](#) and [Eck, D. J.](#) (2021). *International Journal of Forecasting*, 37(4), 1710–1727. doi:10.1016/j.ijforecast.2021.03.010.
- [4] **The Application of PROMETHEE Multi-criteria Decision Aid in Financial Decision Making: Case of Distress Prediction Models Evaluation.**  
[Mousavi, M. M.](#) and [Lin, J.](#) (2020). *Expert Systems with Applications*. doi:10.1016/j.eswa.2020.113438.
- [5] **Identifying Opinion Leaders in Twitter during Social Events: Social Impact Theory from Influencer's Perspective.**  
[Lin, J.](#), [Wang, Z.](#), [Huang, Y.](#), and [Chen, R.](#) (2020). *2019 4th International Conference on Communication and Information Systems*, 197–205. doi:10.1109/ICCIS49662.2019.00042.

<sup>†</sup>Corresponding author.

### Under Review or Revision

- [6] **Distributed Inference for Spatial Quantile Regression.**  
[Lin, J.](#), [Wang, H.](#), and [Wang, L.](#) **Reject & Resubmit**; *Journal of the American Statistical Association: Theory and Methods (JASA T&M)*.
- [7] **Transfer Learning for Survival-based Clustering of Predictors with an Application to TP53 Mutation Annotation.**  
[Liu, X.](#), [Lin, J.](#), [Yan, H.](#), [Shi, H.](#), [Montellier, E.](#), [Chi, E. C.](#), [Hainaut, P.](#), and [Wang, W.](#) **Major revision**; *Journal of the American Statistical Association: Applications and Case Studies (JASA A&CS)*.
- [8] **Censored Bent-line Quantile Regression with Application in a Study on Experimental Autoimmune Myasthenia Gravis.**  
[Lin, J.](#), [Wang, H.](#), and [Luo, J.](#) **Major revision**; *Annals of Applied Statistics (AoAS)*.
- [9] **Striational Antibody-Associated Myositis–Bridging the Gap between Thymoma and Myasthenia Gravis: A Systematic Review.**

Luo, J., Lin, J., Shymansky, J., and Wang, H. J. **Submitted**; *Nature Communications*. [Code](#).

### Working Papers

- [10] **Shed Acetylcholine Receptors from Damaged Muscles Trigger Myasthenia Gravis in a Rodent Model.**  
Lawson, B. S., Li, X., Lin, J., Kuzma, L. C., Pham, A., Wu, Y., Borbi, A., Xu, Q., Wang, H., Garden, O. A., and Luo, J. To be submitted.
- [11] **Quantile Regression for Spatial Survival Data.**  
Lin, J. and Wang, H. J. Work in progress.
- [12] **Predicting Drinking Water Contaminants in Underserved Areas Using Transfer Learning: An Informative Subset-Based Multitask Quantile Deep Neural Network Approach.**  
Liddle, J. L., Lin, J., Wei, E., Shea, K., Zhang, X., and Hu, X. Work in progress.

### Software

---

|              |                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>SQBiT</b> | R package implementing smoothed quantile regression for spatial data. <a href="#">jileil2/SQBiT</a> .                                  |
| <b>DISQ</b>  | Distributed inference framework for spatial quantile regression. <a href="#">jileil2/DISQ</a> .                                        |
| <b>CBQR</b>  | Software for censored bent-line quantile regression. <a href="#">jileil2/cbqr</a> .                                                    |
| <b>PFAS</b>  | Interactive Shiny application for screening post-hurricane private well contamination risk. <a href="#">PFAS Risk Screening Tool</a> . |

### Professional Experience

---

|           |                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025–2026 | <b>Research Assistant</b> , Predicting Drinking Water Contaminants in Underserved Areas with Transfer Learning. Principal investigators: Xindi Hu and Xiaoke Zhang.                |
| 2025      | <b>Research Assistant</b> , Data-Driven Solutions to Mitigate the Impact of Hurricane on Private Well, Drinking Water Quality, and Human Health. Principal investigator: Xindi Hu. |
| 2024–2025 | <b>Research Assistant</b> , Unlocking Complex Heterogeneity in Large-Scale Spatial-Temporal Data with Adaptive Quantile Learning. Principal investigator: Huixia Judy Wang.        |
| 2023–2024 | <b>Research Assistant</b> , Antigen-Specific Immunosuppression of Myasthenia Gravis by CAR-Engineered Tregs. Principal investigator: Jie Luo.                                      |

### Teaching Experience

---

|      |                                                                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Teaching Assistant, Introduction to Mathematical Statistics II (STAT 4158), Introduction to Business and Economic Statistics (STAT 1051), Mathematical Statistics II (STAT 6202), GWU. |
| 2023 | Teaching Assistant, Regression Analysis (STAT 2118), Survival Analysis (STAT 6227), Introduction to Business and Economic Statistics (STAT 1051), GWU.                                 |
| 2022 | Teaching Assistant, Data Analysis (STAT 6210), Mathematical Statistics I (STAT 6201), GWU.                                                                                             |
| 2021 | Teaching Assistant, Advanced Regression Analysis (STAT 527), University of Illinois Urbana-Champaign.                                                                                  |
| 2020 | Teaching Assistant, Applied Regression and Design (STAT 425), University of Illinois Urbana-Champaign.                                                                                 |

### Presentations

---

|      |                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------|
| 2025 | The Annual ASA/IMS Spring Research Conference on Statistics in Industry and Technology, New York, NY. |
| 2025 | The Past, Present and Future of Statistics in the Era of AI, Washington, D.C.                         |
| 2024 | ICSA Applied Statistics Symposium, Nashville, TN.                                                     |
| 2024 | Statistics in the Age of AI, Washington, D.C.                                                         |
| 2023 | ICSA Applied Statistics Symposium, Ann Arbor, MI.                                                     |
| 2018 | INFORMS International Conference, Taipei, Taiwan.                                                     |

### Mentoring



## Honors and Awards

---

|      |                                                                                                |
|------|------------------------------------------------------------------------------------------------|
| 2026 | Minna Miran Kullback Memorial Prize for Leadership and Service, Department of Statistics, GWU. |
| 2025 | Ph.D. Student Poster Competition Award, StatConnect 2025, Vienna, VA.                          |
| 2024 | GW Statistics Cornfield Travel Award, Department of Statistics, GWU.                           |

## Skills

---

|                    |                                                                        |
|--------------------|------------------------------------------------------------------------|
| Programming        | R, Python, Bash, LaTeX.                                                |
| HPC and Automation | SLURM job scripting, GPU/CPU cluster workflows, environment modules.   |
| Reproducibility    | Git/GitHub project management, version control, workflow organization. |

## References

---

|                  |                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Huixia Judy Wang | Professor and Chair, Department of Statistics, Rice University. <a href="mailto:jw322@rice.edu">jw322@rice.edu</a> .                    |
| Feifang Hu       | Professor and Chair, Department of Statistics, The George Washington University. <a href="mailto:feifang@gwu.edu">feifang@gwu.edu</a> . |
| Lily Wang        | Professor, Department of Statistics, George Mason University. <a href="mailto:lwang41@gmu.edu">lwang41@gmu.edu</a> .                    |